

META ADS CAMPAIGN PERFORMANCE TRACKER

Product Requirements Document

Internal multi-tenant reporting & intelligence platform for Meta (Facebook) Ads — built for real estate marketing operations, architected for white-label expansion.

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Table of Contents

Revision Note — v4.1

1. Executive Summary
2. Problem Statement
3. Objectives
4. Success Metrics
5. Assumptions & Constraints
6. Out of Scope / Non-Goals
7. User Roles
8. User Stories
9. Functional Requirements — Modules
10. Non-Functional Requirements
11. Dashboard Specifications
12. Screen-by-Screen Requirements
13. KPI Framework
14. Filters
15. Reports
16. Visualizations
17. Alerts & Notification Rules
18. Data Model & ER Diagram
19. Database Design
20. API Design
21. Role-Based Access Control (RBAC) Matrix
22. UI/UX Guidelines
23. Technical Architecture
24. MVP vs. Phase 2 Roadmap
25. Rollout / Migration Plan
26. Risks
27. Glossary
28. Acceptance Criteria (MVP)
29. Appendix

Revision Note — v4.1

This revision updates the v4.0 draft (July 28, 2026) to reflect two changes requested during review:

- The persona/role set has been corrected to match the roles actually implemented in the live codebase's `workspace_role` enum (admin, manager, marketing_executive, viewer) — the CEO / Business Owner and Marketing Director personas from the v4.0 draft did not exist in the build and have been removed, along with every dashboard, RBAC row, report recipient, and user story tied to them.
- Every module in this document has been cross-checked against the current PulseCast codebase (Next.js 14 App Router + Supabase, project `ghtonvmmwdpcwzbyegnc`) so the PRD describes the system that actually exists plus what's still to be built, not an idealized org chart.

Assumption flagged for review: The codebase defines four roles by name and by which actions require the admin role (invite/remove members, change roles, update workspace settings), but does not yet document the full permission boundary for manager vs. marketing_executive vs. viewer beyond that. Section 21 (RBAC Matrix) below is Claude's best-fit mapping of the old five-persona permissions onto the four real roles — confirm it matches intent before it's treated as spec.

1. Executive Summary

This document specifies a purpose-built, internal reporting and analytics platform for monitoring Meta (Facebook) Ads campaign performance across one Business Manager and multiple Ad Accounts, initially serving a real estate marketing team of 10–25 people, and architected from day one as a multi-tenant (workspace-based) system so it can be white-labelled to other clients later.

The system is explicitly a monitoring, analysis, comparison, and reporting tool — it does not create, edit, or manage live ad campaigns. It becomes the single source of truth that answers, at a glance, what the team actually asks every day: where money is going, which campaigns are working, which need attention, and what to do next.

The platform is designed to be built in a focused MVP first — core campaign tracking, lead analytics, role-based dashboards, and a rule-based Campaign Profitability & Continue/Pause Advisor — with deeper analytics (predictive pacing, lead-quality scoring, natural-language AI narratives) deferred to later phases. This phased approach keeps the initial build scoped to what the team needs on day one, while the underlying architecture is designed to support the full feature set as it's phased in.

2. Problem Statement

Today, campaign performance data lives natively inside Meta Ads Manager, which is built for media buyers running campaigns — not for a small team asking business questions. There is no single view that connects ad spend to property or city performance, and no historical, comparative, or plain "is this working" reporting layer.

Core pain points this system resolves

- No unified view across 300+ campaigns spanning multiple ad accounts under one Business Manager.
- The team cannot quickly answer "what changed since yesterday" or "which campaign needs attention now."
- No systematic way to attribute spend and leads to property or city.
- No standardized daily/weekly/monthly reporting — reporting today is manual and inconsistent.
- No alerting — underperformance (CTR drop, budget exhaustion, learning-limited status) is discovered late, not proactively.
- No defined, repeatable way to tell whether a specific campaign is actually profitable and worth continuing — pause/continue calls are made on gut feel, not a threshold anyone agreed on in advance.
- Revenue/ROI data is partial (deal closures aren't always tracked against ad spend), so true ROAS is currently unmeasurable — the system needs to support an estimated-ROI model using assumed conversion rate and average deal value until full CRM integration exists.

3. Objectives

Each objective below states the business question it answers and why that matters — this is intentional: every module in this PRD traces back to one of these questions, so nothing is built without a clear business purpose.

Objective	Business Question Answered	Why It Matters
Real-time spend visibility	How much are we spending today, across all accounts?	Prevents budget overruns and surprises at month-end billing.
Profitability tracking	Which campaigns are profitable (or trending toward it)?	Redirects budget toward what works, away from what doesn't.

Objective	Business Question Answered	Why It Matters
Account health monitoring	Which ad accounts are underperforming?	Catches account-level issues (policy flags, low delivery) before they compound.
Attention triage	Which campaign needs immediate attention?	Turns a reactive team into a proactive one — issues are surfaced, not discovered.
Profitability & pause guidance	Is this campaign making or losing money, and should it keep running?	Turns raw spend/lead data into a direct continue-or-pause call, without requiring manual analysis every time.
ROI visibility	What is today's (estimated) ROI?	Connects ad spend to business outcomes, not just vanity ad metrics.
Property intelligence	Which property generated the most leads?	Informs where to invest marketing budget across the portfolio.
Geographic intelligence	Which city performs best?	Guides geographic expansion and city-level budget allocation.
Creative intelligence	Which creative performs best?	Reduces creative fatigue and guides the creative team's next brief.
Audience intelligence	Which audience performs best?	Improves targeting efficiency and lowers cost per lead over time.
Change detection	What changed compared to yesterday?	Day-over-day deltas are often more actionable than absolute numbers.
Budget action	Which campaigns should be paused or get more budget?	Turns analysis into a direct, actionable recommendation.

4. Success Metrics

How we'll know the MVP succeeded, 60–90 days after launch:

- 100% of active campaigns (300+) syncing daily with zero manual data entry.
- Daily/weekly reports generated and delivered automatically to each role, replacing manual reporting.
- Time-to-detect a critical issue (e.g. budget exhausted, CTR collapse) drops from "discovered days later" to under 2 hours via alerts.
- At least 80% of active users log in weekly without being reminded — indicating the dashboards are genuinely useful, not just built.
- The team can answer any of the 12 objective questions above in under 60 seconds using the dashboard, without asking a team member to pull a manual report.

5. Assumptions & Constraints

5.1 Assumptions

- One Facebook Business Manager houses multiple Ad Accounts at launch; additional Business Managers (for future white-label clients) will map to additional workspaces.
- Campaign volume at launch is 300+ active campaigns, scaling over time — the data model and sync jobs must be designed for this volume from day one, not retrofitted.

- Lead destination is mixed: some campaigns use native Meta Lead Forms (leads retrievable via Meta API), others send traffic to landing pages/websites (leads arrive via pixel/CRM webhook, not the Meta API directly). The system must support both ingestion paths.
- True revenue/ROAS data is only partially available. Until full CRM integration exists, ROI will be computed using an editable assumed conversion rate and average deal value per property/campaign type, clearly labeled as "Estimated ROI" everywhere it's shown — never presented as confirmed revenue.
- No strict campaign naming convention exists today, and the system will not attempt to auto-parse campaign names to extract property, city, or any other attribute — tagging is manual, in-app, done once per campaign by whoever sets it up.
- The team runs on a flat model: four roles exist in the system (Admin, Manager, Marketing Executive, Viewer), all workspace-scoped rather than campaign-scoped — no user's view is restricted to a subset of campaigns within a workspace they belong to.

5.2 Constraints

- Subject to Meta Marketing API rate limits and data retention windows (Meta API insight data is generally reliable in detail for ~37 months, with some breakdowns limited further).
- Meta attribution windows and platform-reported metrics won't perfectly reconcile with CRM-reported leads/sales — the system should surface both and be transparent about the gap, not force a false reconciliation.
- WhatsApp Business API (for alerting) requires an approved WhatsApp Business Account and message templates pre-approved by Meta — this is a dependency with lead time, not an instant integration.
- Multi-tenant white-labeling later implies workspace-level data isolation must be correct from day one (not bolted on) — this drives the Postgres Row-Level-Security-based architecture already in place (Section 19/23).

6. Out of Scope / Non-Goals

Explicitly excluded from this system, to prevent scope creep during development:

- Creating, editing, launching, or pausing live ad campaigns (this is a read-only reporting layer over Meta's Ads Manager, not a replacement for it).
- Creative design or asset production tools (the system displays creative performance, not a creative editor).
- Full CRM functionality (lead assignment, sales pipeline management, calling) — the system consumes lead/CRM data for analytics, it does not replace a CRM like Ashiyana.
- Ad platforms other than Meta (Google Ads, LinkedIn Ads, etc.) at launch — the architecture should not preclude this later, but it is not built now.
- Automated bidding, budget changes, or campaign actions triggered by the system (all recommendations are advisory; a human takes the action in Ads Manager).
- Payment processing / billing for white-label clients — this PRD covers the product itself, not the commercial SaaS billing layer (that would be a separate workstream once white-labeling is prioritized).

7. User Roles

Four roles cover this system — this matches the `workspace_role` enum already implemented in the database (admin, manager, marketing_executive, viewer). The v4.0 draft of this PRD specified five business personas including a CEO/Business Owner and a Marketing Director; neither role exists in the codebase, and the build has never had a separate executive-oversight layer above Manager. This section replaces that persona set with the four roles that are actually enforced by Row-Level Security today.

Note: Role permissions below (view/edit/delete/export/admin) are Claude's best-fit mapping of business need onto the four implemented roles, informed by the RLS policies already in place (only admin can invite, remove, or re-role members, and only admin can update workspace settings). Confirm the Manager vs. Marketing Executive boundary matches your intent — that split is not yet encoded in application-layer permission checks beyond the membership table itself.

7.1 Admin (Workspace Admin)

Purpose: Give one person the ability to configure the system itself — users, roles, integrations — without needing engineering involvement for routine changes.

Advantage: Makes the system self-serve for onboarding a new workspace/client, which is essential once white-labeling begins.

Attribute	Detail
Goals	Keep the system correctly configured, secure, and connected to the right Meta assets.
Responsibilities	Manage users and roles; connect/reconnect Meta Business Manager and ad accounts; configure alert thresholds and notification channels; review audit logs.
Key KPIs Watched	System health metrics: sync success rate, API error rate, active user count.
Dashboards	Admin Panel (Section 20), plus full business dashboards if also holding a business role.
Permissions	Full system access: user management, role management, integration settings, audit log access, workspace settings. Also has full read/edit/export across business data (enforced today via <code>has_workspace_role(workspace_id, ['admin'])</code> in RLS).
Reports Received	System Health Report (weekly).

7.2 Manager

Purpose: Give the person running day-to-day campaign strategy and owning outcomes end-to-end the operational detail and portfolio-wide visibility needed to act, not just observe.

Advantage: Cuts the time spent manually pulling Ads Manager exports before every status meeting, and lets one person spot cross-account patterns that would be invisible looking account-by-account.

Attribute	Detail
Goals	Optimize budget allocation across accounts and properties; hit lead/CPL targets across the workspace; decide which campaigns to keep funding.
Responsibilities	Own the daily/weekly review cadence; set and adjust targets; act on Continue/Pause recommendations from the Profitability Advisor (Section 9.10); daily campaign monitoring across the full workspace.
Key KPIs Watched	Estimated ROI, CPL, CTR, budget utilization, learning-limited campaigns, Estimated Profit/Loss, Continue/Pause status, account health.
Dashboards	Manager Dashboard (Section 11.1).

Attribute	Detail
Permissions	Full read/edit across the workspace's ad accounts and campaigns; export; approve budget-shift and alert-threshold changes. No user/role management or workspace settings access.
Reports Received	Daily Digest, Weekly Performance Report, Monthly Summary.

7.3 Marketing Executive

Purpose: Give the person executing day-to-day campaign tagging and monitoring the tools to keep data clean and flag issues, without full budget/target authority.

Advantage: Lets the Manager delegate routine monitoring and tagging work without handing over budget-approval or workspace-configuration authority.

Attribute	Detail
Goals	Keep campaign tagging (Property/City) current; catch and flag underperforming or flagged campaigns quickly.
Responsibilities	Daily campaign monitoring; bulk-tag campaigns to Property/City (Section 9.2); acknowledge and triage alerts; escalate loss-making campaigns to the Manager rather than acting unilaterally on Pause recommendations.
Key KPIs Watched	CPL, CTR, delivery/learning status, alert queue, Continue/Pause flags.
Dashboards	Marketing Executive Dashboard (Section 11.2).
Permissions	Read across the workspace; edit tags only (Property/City assignment, alert acknowledge/resolve); export. No edit access to targets, thresholds, or workspace settings; no user management.
Reports Received	Daily Digest.

7.4 Viewer

Purpose: Give anyone who needs to see performance data — a stakeholder, a client-side contact once white-labeled, or a new team member — safe read-only access.

Advantage: Lets the workspace share visibility broadly without risking accidental edits to tags, targets, or thresholds; this is also the default role a newly-invited member receives until an Admin assigns something higher.

Attribute	Detail
Goals	Stay informed on marketing performance without needing to action anything in the system.
Responsibilities	None — this is a passive, informational role.
Key KPIs Watched	Whatever the workspace's dashboards surface; no restricted subset.
Dashboards	Read-only view of the Manager Dashboard's KPI set.
Permissions	Read-only across the workspace. No edit, no delete, no user management. Export access is workspace-configurable (off by default, per the principle of least privilege for the default role).
Reports Received	None by default — a Viewer can be manually added to a report's recipient list by a Manager or Admin.

8. User Stories

A representative set per role — the full backlog will expand these during sprint planning, but every functional requirement in Section 9 traces back to one of these.

Role	User Story
Manager	As a Manager, I want a single number for today's estimated ROI across all accounts, so I don't have to ask anyone for a status update.
Manager	As a Manager, I want to see which of our cities is underperforming this week vs. last week, so I can reallocate budget before month-end.
Marketing Executive	As a Marketing Executive, I want to see which campaigns are Learning Limited across the whole workspace, so I can flag them before they waste further spend.
Marketing Executive	As a Marketing Executive, I want the system to flag which campaigns are losing money and recommend pausing them, so I don't have to manually work out profitability for 300+ campaigns.
Viewer	As a Viewer, I want to see this week's spend and lead totals without being able to change anything, so I can stay informed without risking an accidental edit.
Admin	As an Admin, I want to add a new Ad Account to an existing workspace in under 5 minutes, so onboarding a new client doesn't need engineering.

9. Functional Requirements — Modules

Every module below states its Purpose (why we're building it) and Advantage (what it unlocks), followed by concrete requirements, and is tagged MVP or Phase 2 per the phasing decision in Section 24. Build-status notes (in brackets) reflect what's already live in the codebase as of this revision.

9.1 Dashboard (role-based landing page) [MVP — in progress]

Purpose: Give every role a single first screen that answers their top questions without clicking anywhere.

Advantage: Reduces time-to-insight from minutes of manual filtering to seconds on login.

- Role-aware landing page — content differs by role (Section 11): Admin, Manager, Marketing Executive, Viewer.
- Today vs Yesterday delta on headline KPIs (spend, leads, CPL).
- Top 3 alerts requiring attention, pinned at the top.
- Quick filters: date range, account, property.

9.2 Campaign Monitoring [MVP]

Purpose: Give a live, sortable list of every campaign with health status at a glance.

Advantage: Replaces manually scrolling Ads Manager's 300+ campaign list with a sortable, filterable, taggable table.

- Table view: campaign name, account, status, objective, spend, results, CPL/CPA, delivery status, learning status.
- Health indicator (Green/Amber/Red) computed from CTR, frequency, and pacing thresholds.
- Bulk tagging: manually assign Property and City to campaigns, in bulk or one at a time — no automatic parsing of campaign names (Manager and Marketing Executive can tag; Viewer cannot).
- Search and multi-filter combination (see Section 14).

9.3 Campaign Performance (detail view) [MVP]

Purpose: Give a single campaign's full story — trend, ad sets, ads, creatives — in one drill-down screen.

Advantage: Removes the need to jump between Ads Manager tabs to understand one campaign's full picture.

- Time-series trend chart for spend, CTR, CPL over the selected date range.
- Ad set and ad-level breakdown table.
- Creative thumbnails with performance overlay.
- Day-over-day and week-over-week comparison toggle.

9.4 Lead Analytics [MVP]

Purpose: Connect ad spend to the leads it actually produced, regardless of whether they came via Meta Lead Forms or a landing page.

Advantage: This is the single most requested business answer ("which campaign generated maximum leads") — without it, spend and outcome stay disconnected.

- Unified lead table merging Meta Lead Ads data and landing-page/CRM-sourced leads (dual ingestion per Section 5.1).

- Lead volume trend by campaign, property, city, source.
- Cost per Lead (CPL) computed per campaign/property/city.
- Lead quality tagging field (Phase 2: auto-scored; MVP: manual tag synced from CRM where available).

9.5 Property Analytics [MVP]

Purpose: Roll campaign performance up to the property level — the unit real estate leadership actually thinks in.

Advantage: Lets the team answer "which property generated maximum leads" directly, instead of mentally aggregating campaign names.

- Property-level summary: total spend, leads, CPL, estimated ROI.
- Property comparison view (side-by-side, up to 5 properties).
- Property leaderboard, sortable by any KPI.

9.6 Audience Analytics [Phase 2]

Purpose: Show which audience segments are actually converting, not just which have the most reach.

Advantage: Improves targeting efficiency over time by making audience fatigue and winners visible, not buried in Ads Manager breakdowns.

- Audience performance table: age, gender, placement, device breakdowns.
- Audience overlap / fatigue indicator.
- Best-performing audience recommendation per property type.

9.7 Creative Analytics [Phase 2]

Purpose: Show which ad creative (image/video/copy) is actually driving results, and flag fatigue before it hurts CPL.

Advantage: Gives the creative team a data-backed brief for what to make next, instead of guessing.

- Creative-level performance table with thumbnails: CTR, hook rate, cost per view, CPL.
- Creative fatigue alert (frequency rising + CTR declining).
- Creative leaderboard by property/campaign objective.

9.8 Budget Tracking & Pacing [MVP]

Purpose: Show, in real time, whether each campaign/account is on pace to over- or under-spend its budget.

Advantage: Prevents the two most common budget failures: running out mid-month, or under-spending an approved budget.

- Budget utilization % and pacing indicator (ahead/on-track/behind) per campaign and account.
- Forecast spend (simple linear projection at MVP; Phase 2: seasonality-aware forecast).
- Remaining budget and days-until-exhaustion estimate.

9.9 Daily / Weekly / Monthly Reports [MVP]

Purpose: Automatically generate the reports people currently build by hand.

Advantage: Frees the team from manual report-building and guarantees consistent, comparable reporting every period.

- Auto-generated Daily Digest (per role) delivered via email + WhatsApp + in-app.
- Weekly Performance Report (Monday morning, prior week summary).
- Monthly Summary (first business day of the month).
- Phase 2: Quarterly/Yearly and fully custom date-range reports, scheduled report builder.

9.10 Campaign Profitability & Continue/Pause Advisor [MVP]

Purpose: *Turn spend and lead data into a direct, per-campaign answer to "is this making money, and should it keep running" — computed the same way every time, not eyeballed.*

Advantage: *Replaces gut-feel pause/continue calls with a repeatable, auditable rule, and surfaces loss-making campaigns before more budget is wasted on them.*

- Per-campaign inputs: Spend (to date and trailing 7-day), Cost per Lead, Estimated Revenue (assumed conversion % $\times$ average deal value, per Section 5.1), Estimated Profit/Loss = Estimated Revenue – Spend, and trend direction over the trailing 7/14 days.
- Deterministic rule-based classification per campaign: Profitable / Break-even / Loss-making, against workspace-configurable thresholds (e.g. Estimated Profit/Loss margin, days spent below break-even).
- Recommendation output: Continue / Monitor / Reduce Budget / Pause, each paired with a plain-language reason built from the specific rule that fired — not an AI-generated explanation, a templated one traceable to the exact numbers.
- Recommendation is advisory only — no automatic pausing; a human acts on it in Ads Manager, consistent with Section 6 (Out of Scope).
- Thresholds are configurable per workspace by an Admin, not hardcoded.
- Surfaced on the Campaign Monitoring table (Section 9.2), Campaign Detail view (Section 9.3), and as a dedicated widget on the Manager and Marketing Executive dashboards (Section 11). No external AI/LLM API dependency.

9.11 Alerts & Notifications [MVP]

Purpose: *Proactively surface problems the moment they cross a defined threshold, instead of waiting for someone to notice.*

Advantage: *Converts the system from a passive report into an active early-warning system — this is the highest-leverage module in the whole platform.*

- Rule engine per Section 17, configurable thresholds per workspace.
- Delivery via Email, WhatsApp Business API, and in-app push (PWA) as decided in Section 5.
- Acknowledge / Resolve / Escalate workflow with audit trail.

9.12 Export Centre [MVP]

Purpose: *Let any authorized user get their data out, in the format they need, without asking engineering.*

Advantage: *Removes a common support-ticket category ("can you send me this as Excel") entirely.*

- Export to CSV/XLSX for any table view, respecting current filters.
- PDF export for report views (branded, workspace-logo aware for white-label).
- Export history/audit log.

9.13 Admin & Role Management [MVP — live]

Purpose: *Let an Admin manage users, roles, and permissions without engineering involvement.*

Advantage: *Essential for onboarding new workspaces quickly once white-labeling starts.*

- User invite/deactivate flow (workspace_members table, admin-only insert/update/delete per current RLS policy).
- Role assignment per the RBAC matrix (Section 21) — role stored as the workspace_role enum: admin, manager, marketing_executive, viewer.
- Workspace-level settings: branding (white-label), timezone, currency, alert thresholds.

9.14 Meta API Integration [MVP — live]

Purpose: *Reliably pull campaign, ad set, ad, and insight data from Meta's Marketing API on a schedule.*

Advantage: *This is the data foundation everything else depends on — without reliable sync, every dashboard is stale or wrong.*

- System User-based authentication per workspace — implemented as the pulsecastsyncbot System User on the "PulseCast Ads Sync" Facebook app (Employee-level access, not a personal login).
- Daily scheduled sync via Vercel Cron, plus on-demand manual refresh.
- Sync status visibility (last synced timestamp, error state) surfaced to Admins.
- Graceful handling of Meta API rate limits, token expiry, and partial failures (retry with backoff).

9.15 Audit Logs [MVP]

Purpose: *Record who did what, when — every settings change, alert resolution, and export.*

Advantage: *Required for accountability once multiple people and multiple client workspaces share the system, and for diagnosing "who changed this threshold" disputes.*

- Immutable log of user actions: logins, exports, settings changes, alert acknowledgments, role changes.
- Admin-only searchable log view with date/user/action filters.

9.16 Settings / Workspace Management [MVP — live]

Purpose: *Let each workspace (client) configure its own accounts, users, and preferences independently.*

Advantage: *This is what makes the multi-tenant architecture actually usable by non-technical admins, not just correct in the database.*

- Workspace switcher for users who belong to multiple workspaces (e.g. the founder across real estate + future clients).
- Per-workspace: connected ad accounts, currency, timezone, branding, alert thresholds, notification channel preferences.

10. Non-Functional Requirements

Category	Requirement
Performance	Dashboard initial load under 2s for cached data; filtered queries under 3s at 300+ campaign scale.
Scalability	Data model and sync jobs must handle 10x campaign volume (3,000+) and multiple workspaces without redesign.
Caching	Aggregate KPI rollups pre-computed on sync (not calculated live on every dashboard load).
Offline support (PWA)	Last-synced dashboard data viewable offline; clear "stale data" indicator when offline.
Accessibility	WCAG 2.1 AA color contrast on all charts and KPI cards; keyboard-navigable tables.
Security	Row-Level Security enforced at the database layer per workspace (not just application-layer checks) — implemented via <code>has_workspace_role()</code> and <code>is_workspace_member()</code> ; encrypted secrets for Meta API tokens.
Auditability	Every settings change and data export logged (Section 9.15).
Availability	Target 99.5% uptime for the dashboard; sync jobs must alert Admins on failure, not fail silently.
Error handling	User-facing errors are plain-language; technical detail logged, not shown to end users.
Monitoring	Sync job success/failure, API error rates, and page load times tracked and visible to Admins.

11. Dashboard Specifications

Three business dashboards cover the four roles (Admin has its own configuration-focused Admin Panel per Section 9.13, plus full read access to whichever business dashboard it needs).

11.1 Manager Dashboard

Purpose: *Give the person who owns marketing outcomes the full portfolio view with drill-down into any dimension.*

Advantage: *Lets one person manage 300+ campaigns' worth of complexity without needing to open Ads Manager.*

Widgets / Cards: Spend & leads trend (with day-over-day delta), Estimated ROI, Estimated Revenue, Top 3 / Bottom 3 properties, Account health heatmap, Property leaderboard, City leaderboard, Profitability & Continue/Pause recommendation panel (Section 9.10), Budget pacing table, Alert panel, Target achievement gauge (daily/weekly/monthly).

Filters & Drill-down: Date range, account, property, city, workspace (if multi-client access). Drill-down from any chart into Campaign Monitoring.

11.2 Marketing Executive Dashboard

Purpose: *Give a Marketing Executive everything needed to run the full campaign portfolio day-to-day.*

Advantage: *Turns the daily routine from a manual Ads Manager sweep into a 5-minute dashboard review.*

Widgets / Cards: Spend/leads/CPL today vs yesterday (full workspace), Campaigns needing attention (Red/Amber), Continue/Pause recommendations (Section 9.10) with reason, Budget pacing, Alert panel.

Filters & Drill-down: Date range, account, property. Drill-down to Campaign Performance detail.

11.3 Viewer Dashboard

Purpose: *Give a Viewer the same picture of performance as everyone else, without any control that could change data.*

Advantage: *Lets the workspace share visibility broadly — a client contact, a new hire, a stakeholder — without any risk to campaign tags, targets, or thresholds.*

Widgets / Cards: Read-only version of the Manager Dashboard's KPI cards and trend charts; no recommendation-panel actions, no threshold configuration.

Filters & Drill-down: Date range, account, property, city — view only; no bulk-tagging, no export unless the workspace has enabled Viewer export.

12. Screen-by-Screen Requirements

Screen	Key Requirements
Login / Workspace Switcher	Email/password or SSO login; if user belongs to >1 workspace, workspace picker shown before dashboard.
Dashboard (role-based)	As specified per role in Section 11; loads in under 2s from cache.
Campaign Monitoring (list)	Sortable/filterable table, health indicators, bulk tagging (Manager/Marketing Executive), saved filter views.
Campaign Detail	Trend charts, ad set/ad breakdown, creative thumbnails, comparison toggle.
Lead Analytics	Unified lead table, source breakdown, CPL by dimension, quality tag field.
Property Analytics	Property summary cards, comparison view, leaderboard.
Profitability & Recommendation View	Per-campaign Profit/Loss classification, Continue/Pause recommendation with reason, threshold configuration (Admin only).
Reports Centre	List of generated reports (daily/weekly/monthly), download/resend, Phase 2 custom report builder.
Alerts Centre	Full alert feed, filter by severity/status, acknowledge/resolve/escalate actions.
Export Centre	Export history, format selector, filter-aware export trigger.
Admin Panel	User management, role assignment, workspace settings, connected ad accounts, audit log.

13. KPI Framework

KPIs are grouped by category rather than listed flat — this makes the (very long) list from the original brief usable in a UI, since each group maps naturally to a filter tab or dashboard section.

13.1 Delivery & Reach

KPI	Definition / Note
Spend	Total amount spent, by campaign/account/day.
Budget & Budget Utilization %	Configured budget vs. amount spent so far.
Impressions	Number of times an ad was shown.
Reach	Number of unique people who saw an ad.
Frequency	$\text{Impressions} \div \text{Reach}$ — high frequency signals ad fatigue.
CPM	Cost per 1,000 impressions.
Delivery Status / Learning Limited	Meta's own delivery health signal, surfaced directly, not recalculated.

13.2 Engagement & Clicks

KPI	Definition / Note
CTR / Unique CTR	Click-through rate, overall and per unique user.
CPC	Cost per click.
Outbound Click / Link Click	Clicks that leave the platform vs. any link click.
Landing Page View	Confirms the click resulted in an actual page load.
Video View / View Rate / Hook Rate	Video-specific engagement depth, key for creative diagnostics.
Post Engagement (Likes, Comments, Shares, Saves)	Organic-style engagement signals on the ad unit.

13.3 Cost Efficiency

KPI	Definition / Note
Cost per Result	Meta's own optimization-event cost metric.
Cost per View / Cost per Purchase	Cost efficiency for video and purchase-objective campaigns.

13.4 Lead & Conversion (core to real estate)

KPI	Definition / Note
Lead / Qualified Lead	Raw lead count vs. CRM-confirmed qualified lead.
CPL (Cost per Lead)	$\text{Spend} \div \text{Leads}$ — the single most-watched KPI for this business.
Conversion Rate	$\text{Leads} \div \text{Clicks}$, or $\text{Qualified Leads} \div \text{Leads}$ depending on the funnel stage viewed.
Lead Quality Score	MVP: manual tag from CRM feedback. Phase 2: modeled score from historical conversion patterns.
Estimated ROAS / Estimated ROI	Computed using assumed conversion % and average deal value per property (Section 5.1) — always labeled "Estimated".

13.5 Budget & Pacing

KPI	Definition / Note
Pacing (Ahead / On-track / Behind)	Spend-to-date vs. expected spend-to-date given the budget period.
Forecast Spend / Forecast Leads	Linear projection at MVP; seasonality-aware in Phase 2.
Remaining Budget	Budget minus spend-to-date.
Daily / Weekly / Monthly Target & Achievement %	User-configured targets vs. actuals.

13.6 Account & Ranking Signals

KPI	Definition / Note
Quality / Engagement / Conversion Ranking	Meta's relative ad-quality diagnostics, surfaced as-is.
Auction Overlap	Flags when your own campaigns compete against each other.

13.7 Profitability & Recommendation (Section 9.10)

KPI	Definition / Note
Estimated Profit/Loss	Estimated Revenue minus Spend for a campaign — the core input to the advisor's classification.
Profitability Classification	Profitable / Break-even / Loss-making, against workspace-configured thresholds.
Recommendation Status	Continue / Monitor / Reduce Budget / Pause — advisory only, always paired with a stated reason.
Days Below Break-even	Consecutive-day counter used by the rule engine before a Pause recommendation triggers.

Phase 2 note: Lead Quality Score and full ROAS become precise once CRM integration is deeper (Section 24); until then they are clearly marked as estimates everywhere they appear in the UI.

14. Filters

Filters are combinable (AND logic across groups, OR within a group — e.g. City: Pune OR Mumbai, AND Recommendation: Pause).

Filter Group	Options
Date	Today, Yesterday, Last 7 Days, Last 30 Days, Custom Range, Week-over-Week / Month-over-Month compare toggle.
Campaign	Campaign, Objective, Status, Buying Type.
Account / Structure	Ad Account, Business Manager (Phase 2, once multi-BM), Platform, Placement, Device.
Geography	Country, State, City.
Business Attribution	Property.
Profitability	Classification (Profitable / Break-even / Loss-making), Recommendation (Continue / Monitor / Reduce Budget / Pause).
Audience & Creative	Audience, Creative, Gender, Age.
Attribution / Tech	Lead Source, Pixel, Event, UTM parameters.

15. Reports

Report	Frequency	Recipients	Phase
Daily Digest	Daily, 9 AM IST	All roles (scoped to their access)	MVP
Weekly Performance Report	Every Monday	Manager	MVP
Monthly Summary	1st business day of month	Manager, Admin	MVP
Property Performance Report	Weekly	Manager	MVP
Campaign Performance Report	On-demand	Manager, Marketing Executive	MVP
Profitability & Recommendation Report	Weekly	Manager, Marketing Executive	MVP
Creative Performance Report	Weekly	Marketing Executive, Viewer	Phase 2
Audience Performance Report	Weekly	Viewer	Phase 2
Lead Quality / Sales Funnel Report	Weekly	Manager	Phase 2 (needs deeper CRM link)
Quarterly / Yearly / Custom Report Builder	On-demand	All (permission-gated)	Phase 2

16. Visualizations

Chart choice mapped to what each metric type needs to communicate — not chosen for visual variety.

Metric Type	Recommended Chart
Single headline number (spend, ROI today)	KPI Card with delta indicator (▲/▼vs. yesterday).
Trend over time (spend, CPL, leads)	Line Chart / Area Chart.
Comparison across categories (properties, cities)	Bar Chart / Leaderboard.
Composition (spend by objective, leads by source)	Donut Chart (avoid Pie for >4 segments).
Hierarchical composition (spend by account > campaign > ad set)	Treemap.
Two-dimensional health view (campaign x day)	Heatmap.
Geographic performance (city/state)	Geo Map.
Relationship between two metrics (spend vs. leads)	Scatter Plot / Bubble Chart (bubble size = 3rd metric, e.g. CPL).
Funnel-shaped drop-off (impressions > clicks > leads > sales)	Funnel Chart.
Contribution to a total change (why spend changed WoW)	Waterfall Chart.
Target vs. actual (daily/weekly/monthly target)	Gauge Chart.
Ranked list (top properties, top/bottom profitability)	Leaderboard (styled table, not a generic chart).
Multi-dimensional cross-tab (Analyst use)	Pivot Table / Matrix.

17. Alerts & Notification Rules

Alerts are the highest-leverage module (Section 9.11) — this table is the initial rule set; all thresholds are configurable per workspace by an Admin.

Alert	Trigger Condition	Default Severity	Channel
CTR below threshold	CTR < workspace-defined minimum for 2+ consecutive days	Amber	Email + In-app
CPA / CPL increased	CPL rises >25% vs. trailing 7-day average	Amber	Email + In-app
Budget exhausted	Remaining budget = 0 or campaign paused by Meta for budget	Red	WhatsApp + Email + In-app
Campaign rejected	Meta returns ad review rejection status	Red	WhatsApp + Email + In-app
Campaign stopped unexpectedly	Active campaign transitions to Paused/Inactive without a logged reason	Red	WhatsApp + In-app
Learning Limited	Meta reports Learning Limited delivery status	Amber	In-app
Frequency high	Frequency exceeds workspace threshold (e.g. >3.5 in 7 days)	Amber	In-app
Estimated ROAS dropped	Estimated ROI trend declines >20% week-over-week	Amber	Email + In-app
Pause recommended	Profitability Advisor (Section 9.10) classifies a campaign Loss-making beyond the configured consecutive-day threshold	Red	WhatsApp + Email + In-app
Spend anomaly	Day's spend deviates >40% from trailing 7-day average (either direction)	Red	WhatsApp + Email + In-app
Lead volume dropped	Daily leads fall >30% vs. trailing 7-day average	Amber	Email + In-app
Pixel inactive	No pixel events received for 24+ hours on an active campaign	Red	Email + In-app
Sync / API failure	Meta API sync job fails or token expires	Red (Admin-only)	Email + In-app (Admin)

18. Data Model & ER Diagram

Purpose: Define every entity the system tracks and how they relate, before any table is built.

Advantage: Prevents costly schema rework later — entity relationships defined up front catch data-isolation and migration-ordering issues before they're expensive to fix.

18.1 Core Entities

- Workspace (Client) — the multi-tenancy boundary; every other entity below belongs to exactly one Workspace.
- Business Manager — one per Workspace at launch.
- Ad Account — belongs to a Business Manager; multiple per Workspace.
- Campaign — belongs to an Ad Account; tagged with Property, City.
- Ad Set — belongs to a Campaign.
- Ad — belongs to an Ad Set; references a Creative.
- Creative — reusable image/video/copy asset, referenced by one or more Ads.
- Audience — targeting definition, referenced by one or more Ad Sets.
- Pixel — tracking asset linked to an Ad Account, referenced by landing-page-sourced Leads.
- Lead — sourced from Meta Lead Ads or Pixel/CRM webhook; linked to a Campaign and Property.
- Property — the real estate unit leads are attributed to.
- Profitability Snapshot — computed record per Campaign per evaluation run: Estimated Profit/Loss, Classification, Recommendation, and the specific rule/reason that produced it (Section 9.10).
- Customer — optional, populated when CRM integration provides post-lead sales data.
- Daily Metrics — one row per Campaign per day: spend, impressions, clicks, leads, etc. (the core fact table).
- Historical Metrics — same shape as Daily Metrics, retained long-term for trend/YoY analysis.
- Alert — generated instance of a triggered rule from Section 17, linked to a Campaign/Account.
- Notification — delivery record of an Alert or Report to a User via a Channel.
- Audit Log — immutable record of a User action.
- Workspace Member — the RBAC join table already implemented: workspace_id, user_id, and a role column typed as the workspace_role enum (admin, manager, marketing_executive, viewer), unique per (workspace_id, user_id).

18.2 ER Diagram — Provision

An entity-relationship diagram should be produced as a visual artifact alongside this PRD (not embedded as static text) so it stays easy to update as the schema evolves during sprint planning. The core relationships to diagram are:

- Workspace 1—* Business Manager 1—* Ad Account 1—* Campaign 1—* Ad Set 1—* Ad *—1 Creative
- Campaign *—1 Property
- Campaign 1—* Daily Metrics (time-series fact table)
- Campaign 1—* Profitability Snapshot (one per evaluation run, computed from Daily Metrics + Section 5.1 revenue assumptions)
- Campaign 1—* Lead, Lead *—1 Property, Lead *—0..1 Customer (once CRM-linked)
- Ad Account 1—* Pixel, Pixel 1—* Lead (for landing-page-sourced leads)

- Campaign/Account 1—* Alert, Alert 1—* Notification, Notification *—1 User
- Workspace 1—* Workspace Member, Workspace Member *—1 User, Workspace Member has one role (workspace_role enum)

This diagram provision is intentionally left as a to-be-rendered artifact rather than an ASCII sketch — say the word and it can be generated as a visual right after this PRD is finalized.

19. Database Design

Aspect	Recommendation
Primary/Foreign Keys	UUID primary keys throughout (matches Supabase convention); every child table carries a workspace_id foreign key, even where it could be inferred via a join — this is what makes Row-Level Security simple and fast.
Indexes	Composite index on (workspace_id, campaign_id, date) for Daily Metrics — this is the hot query path for every dashboard. Live today: idx_workspace_members_lookup on (workspace_id, user_id) and idx_workspace_members_user on (user_id).
Partitioning	Partition Daily Metrics / Historical Metrics by month once volume grows past ~6-12 months of 300+ campaigns/day — not needed at MVP launch, but the schema should be partition-ready.
Archiving Strategy	Roll data older than 13 months from Daily Metrics into a compressed Historical Metrics table on a monthly job; dashboards query Historical Metrics transparently for older ranges.
Historical Data Strategy	Preserve raw daily granularity for 24 months minimum (real estate sales cycles are long, per background context); aggregate-only beyond that.
Multi-tenancy Enforcement	Postgres Row-Level Security policy on every table keyed to workspace_id, enforced at the database layer via has_workspace_role() / is_workspace_member() helper functions, with a SECURITY DEFINER bootstrapping function for initial workspace/user setup, so a bug in application code cannot leak one workspace's data into another's.
Role Enforcement (live)	workspace_role enum (admin, manager, marketing_executive, viewer) on workspace_members.role. Current RLS: only admin can insert/update/delete workspace_members rows and update workspace settings; all members can select the membership list of workspaces they belong to.

20. API Design

Internal REST API (consumed by the Next.js frontend and any future white-label client apps), all routes scoped under `/api/workspaces/{workspaceId}/...` to keep the multi-tenancy boundary explicit at the URL level.

Domain	Representative Endpoints
Dashboard	GET /dashboard/summary, GET /dashboard/alerts
Campaign	GET /campaigns, GET /campaigns/{id}, PATCH /campaigns/{id}/tags
Metrics	GET /metrics/daily, GET /metrics/compare
Profitability	GET /campaigns/{id}/profitability, GET /profitability/recommendations, PATCH /workspace/profitability-thresholds
Reports	GET /reports, GET /reports/{id}, POST /reports/generate
Filters	GET /filters/options (returns valid filter values for current workspace)
Export	POST /export, GET /export/history
Authentication	POST /auth/login, POST /auth/refresh, POST /auth/logout
Notification	GET /notifications, PATCH /notifications/{id}/read
Alert	GET /alerts, PATCH /alerts/{id}/acknowledge, PATCH /alerts/{id}/resolve
Settings	GET/PATCH /workspace/settings, GET/POST /workspace/users, GET/POST /workspace/ad-accounts

21. Role-Based Access Control (RBAC) Matrix

Reduced to the four roles enforced today by the workspace_role enum and the current RLS policies.

Role	View	Edit (tags/targets)	Delete	Export	Approve	Manage Users/Settings
Admin	All (own workspace)	Yes	Users only	Yes	Yes	Yes
Manager	All (own workspace)	Yes	No	Yes	Yes (budget shifts)	No
Marketing Executive	All (own workspace)	Tags/alerts only	No	Yes	No	No
Viewer	All (own workspace)	No	No	Workspace-configurable	No	No

22. UI/UX Guidelines

Visual and interaction reference points: Power BI and Looker Studio for dashboard density and drill-down patterns; Grafana for alert-panel styling; Meta Ads Manager for table/filter conventions users already know; HubSpot for the overall approachable, non-technical tone appropriate for Manager/Viewer roles.

- Minimal, professional visual language — data density without clutter; generous whitespace around KPI cards.
- Responsive: full desktop dashboard experience, plus a genuinely usable mobile/PWA view for on-the-go checks (this is a PWA by requirement, not an afterthought).
- Fast: skeleton loaders, not blank screens, while data loads; cached last-known values shown instantly, live data streamed in after.
- Dark/Light mode, respecting system preference by default.
- Consistent health-status color coding across the entire app: Green = healthy, Amber = needs attention, Red = critical — used identically on every screen, not redefined per module.
- White-label readiness: logo, primary color, and workspace name are theme variables, not hardcoded, from the first UI component built.

23. Technical Architecture

Purpose: *Recommend a stack that's fast to build, well suited to this scale, and ready for multi-tenancy from day one.*

Advantage: *Choosing a proven, well-documented stack up front reduces build risk and keeps the team from re-deciding architecture mid-project.*

Layer	Recommendation	Status
Frontend	Next.js 14 (App Router), TypeScript, Tailwind CSS	Live
Backend	Next.js API routes / server actions (modular monolith, not microservices)	Live
Database	Supabase (managed Postgres) with Row-Level Security per workspace — project ghtonvmmwdpcwzbyegnc	Live
Caching	Pre-computed daily rollup tables (populated by the sync job) + Next.js data caching	In progress
Authentication	Supabase Auth, workspace-scoped sessions with a workspace switcher	Live
Meta Marketing API Integration	System User pulsecastsyncbot per workspace (Employee role), on the "PulseCast Ads Sync" app (App ID 138703756861852), scheduled via Vercel Cron	Live — 5-campaign sync test passed
Background Jobs / Queue	Vercel Cron for scheduled sync (currently 1:00 AM IST daily); a lightweight queue for alert evaluation, notification dispatch, and Profitability Advisor rule evaluation (Section 9.10)	Sync live; queue not yet built
Profitability Rules Engine	Deterministic threshold logic implemented as a Postgres function or Next.js server-side job — no external AI/LLM API call	Not yet built
Notifications	Email via a transactional provider (e.g. Resend/SendGrid); WhatsApp via WhatsApp Business API (Meta Cloud API) with pre-approved templates; in-app via Supabase Realtime	Not yet built
Hosting	Vercel (frontend + API routes)	Live
CI/CD	GitHub Actions → Vercel preview deployments → production on merge to main	Live

23.1 Current Implementation Note

As of this revision, the following pieces of Section 23's recommended architecture already exist in the live codebase (GitHub: ABSCoder1981) and this PRD has been reconciled against them rather than describing a parallel idealized design:

- workspace_role enum ('admin', 'manager', 'marketing_executive', 'viewer') on the workspace_members join table — this is the single source of truth for Section 7 (Roles) and Section 21 (RBAC Matrix) above.
- RLS policies already enforce: any workspace member can view the membership list; only admins can insert, update, or delete workspace_members rows; only admins can rename or soft-delete a workspace.
- A SECURITY DEFINER bootstrapping function resolves the chicken-and-egg problem of creating the first workspace/admin membership before any RLS policy exists to permit it.
- Workspace-scoped URL structure (/workspaces/[workspaceId]/...) is chosen over session-based workspace tracking.
- Facebook Developer app "PulseCast Ads Sync" and the pulsecastsyncbot System User are live and have completed a successful sync test.

24. MVP vs. Phase 2 Roadmap

MVP covers core tracking, role-based dashboards, and a rule-based profitability advisor; natural-language AI narratives and predictive analytics are deferred.

Phase	Scope	Excludes
MVP (Phase 1)	Dashboard, Campaign Monitoring, Campaign Performance, Lead Analytics, Property Analytics, Budget Tracking, Daily/Weekly/Monthly Reports (auto-generated only), Campaign Profitability & Continue/Pause Advisor (rule-based), Alerts & Notifications, Export Centre, Admin & Role Management, Meta API Integration, Audit Logs, Workspace Settings.	Audience Analytics, Creative Analytics, AI-generated narrative explanations, custom report builder, quarterly/yearly reports, deep CRM/lead-quality scoring.
Phase 2	Audience Analytics, Creative Analytics, Lead Quality scoring (model-based), custom/scheduled Report Builder, seasonality-aware forecasting, deeper CRM integration for true (not estimated) ROI, optional AI-generated plain-English explanations layered on top of the existing rule-based advisor.	—
Phase 3 (Future)	Multi-Business-Manager support for onboarding external white-label clients, billing layer, additional ad platforms (Google Ads), native mobile apps.	—

Sprint order agreed for the MVP build: Meta API + core data model with RLS → Campaign Monitoring → Lead & Property Analytics → Profitability Advisor → Budget Pacing → Alerts (Email/in-app first, WhatsApp deferred) → role-based dashboard variants.

25. Rollout / Migration Plan

- Stage 1 — Connect one Ad Account first (pilot), validate sync accuracy against Ads Manager's own reported numbers before onboarding the remaining accounts.
- Stage 2 — Onboard all remaining Ad Accounts under the single Business Manager; run the system in parallel with any existing manual reporting for one full week to build trust in the numbers.
- Stage 3 — Switch daily/weekly reporting fully to the automated system; retire manual reports.
- Stage 4 — Roll out role-based dashboards, including the Profitability Advisor, to Managers and Marketing Executives with a short walkthrough (no formal training module needed at this team size).
- Stage 5 (later) — When the first white-label client is signed, use that onboarding as the test of the multi-tenant workspace model end-to-end before selling it as a repeatable process.

26. Risks

Risk	Impact	Mitigation
Meta API rate limits / breaking changes	Sync failures, stale dashboards	Backoff/retry logic, sync-status monitoring surfaced to Admins, version-pin API calls.
"Estimated ROI" mistaken for confirmed revenue	Bad budget decisions made on unreliable numbers	Persistent "Estimated" labeling and a visible assumptions panel (conversion %, deal value) wherever ROI is shown.
Untagged campaigns (manual tagging skipped or delayed)	Broken property/city attribution, campaigns missing from property-level rollups	Bulk tagging UI (Section 9.2) makes manual tagging fast; data-quality completeness indicator on the dashboard flags untagged campaigns.
Pause recommendation acted on too literally	A campaign paused on an early or noisy signal, losing legitimate long-term leads	Recommendation is always advisory with a visible reason and minimum-spend/minimum-day thresholds before it can fire (Section 9.10); no auto-pausing.
WhatsApp Business API approval delay	Alert channel unavailable at launch	Launch alerts on Email + In-app first; add WhatsApp once templates are approved, without blocking MVP.
Multi-tenant data leakage across workspaces	Severe — client data exposure once white-labelled	Database-layer Row-Level Security (Section 19), never relying on application-layer checks alone.
Role permissions drift from what's documented here	Manager vs. Marketing Executive boundary implemented inconsistently across features as the build grows	Section 21's RBAC Matrix is the single source of truth; any new feature's permission check should be written against it, and this PRD updated the same day a role's permissions change in code.
Scope creep back toward the full 25-section spec at once	Delayed MVP launch	Phase gating (Section 24) enforced in sprint planning; Phase 2/3 items explicitly deferred, not silently absorbed.

27. Glossary

Term	Definition
CPL	Cost per Lead — total spend divided by number of leads generated.
CPA	Cost per Action/Acquisition — cost per defined conversion event.
CTR	Click-Through Rate — clicks divided by impressions.
CPM	Cost per Mille — cost per 1,000 impressions.
ROAS	Return on Ad Spend — revenue generated divided by ad spend.
ROI	Return on Investment — net profit relative to total investment, broader than ROAS.
Hook Rate	Percentage of viewers who watch the first few seconds of a video ad — an early-engagement signal.
Frequency	Average number of times a unique person has seen an ad within a period.
Learning Limited	A Meta delivery status indicating a campaign lacks enough conversion events to optimize effectively.
Pacing	Whether spend-to-date is ahead of, on track with, or behind the expected rate given the budget period.
Workspace	The multi-tenancy boundary in this system — one client/organization's isolated data and users.
Workspace Role	One of four values in the workspace_role enum — admin, manager, marketing_executive, viewer — assigned per user per workspace via the workspace_members table.
System User	A Meta Business Manager identity used for server-to-server API access, not tied to a personal login. Implemented as pulsecastsyncbot.
RLS (Row-Level Security)	A database-layer security mechanism restricting which rows a query can return based on the requesting user's context (here, their workspace and role).
Break-even Threshold	The Estimated Profit/Loss point at which a campaign is judged neither profitable nor loss-making — the boundary the Profitability Advisor's rules are set against.
Profitability Advisor	The rule-based module (Section 9.10) that classifies each campaign as Profitable / Break-even / Loss-making and recommends Continue / Monitor / Reduce Budget / Pause, with a stated reason.

28. Acceptance Criteria (MVP)

- All active campaigns (300+) sync daily with a visible last-synced timestamp; sync failures generate an Admin alert within 15 minutes.
- Every role's dashboard (Section 11) loads with role-correct data scope, verified against the RBAC matrix (Section 21) — no user sees or edits data outside their permission scope.
- Daily Digest is delivered automatically via Email and In-app (WhatsApp once approved) to every active user without manual triggering.
- CPL, spend, and lead counts on the dashboard reconcile with Meta Ads Manager's own reported numbers within an agreed tolerance (e.g. $\pm 2\%$) during the pilot stage (Section 25, Stage 1).
- Estimated ROI is visibly and consistently labeled "Estimated" everywhere it appears, with the underlying assumption (conversion %, deal value) visible on hover/click.
- An Admin can invite a user, assign one of the four roles, and connect a new Ad Account without engineering assistance.
- All alert rules in Section 17 (including the Profitability Advisor's Pause recommendation) fire correctly against test data and can be acknowledged/resolved by an authorized role.
- The Profitability Advisor's classification and recommendation for a given campaign are reproducible from its stated inputs (Spend, Estimated Revenue, thresholds) — the same numbers always produce the same verdict.
- Row-Level Security is verified to block cross-workspace data access even via a direct API call (not just hidden in the UI) — including verifying that only admin-role members can modify `workspace_members` or workspace settings.

29. Appendix

29.1 Key Design Decisions Log

Decision	Chosen Direction
System scope	New, standalone system — not an extension of any prior project.
Ad account structure	One Business Manager, multiple Ad Accounts at launch.
Client scope	Real estate only at launch; architected to be white-labelled to other clients/verticals later.
Tenancy model	Multi-tenant (workspace-based) from day one, not retrofitted later.
Campaign volume	Designed for 300+ active campaigns at launch.
Lead data sources	Mixed — both native Meta Lead Forms and landing-page/website leads must be supported.
Campaign naming/tagging	No auto-parsing of campaign names — Property/City tagging is manual, in-app, by design (not a naming-convention fallback).
Revenue/ROI data	Partial — ROI is estimated using an editable assumed conversion rate and deal value, always labeled "Estimated."
Alert channels	Email + WhatsApp Business API + in-app push (PWA).
Role model (revised in v4.1)	Four workspace-scoped roles matching the implemented workspace_role enum: Admin, Manager, Marketing Executive, Viewer. The CEO/Business Owner and Marketing Director personas from v4.0 do not exist in the codebase and have been removed from this document.
Org hierarchy scope	Flat model — no per-person campaign attribution; all four roles see the full workspace, differing only in edit/admin permissions, not in which campaigns they can see.
Profitability advisor	Rule-based (deterministic) Profit/Loss classification and Continue/Pause recommendation on spend/CPL/estimated-revenue thresholds; no external AI/LLM API dependency at MVP.
Release strategy	MVP first (core tracking, dashboards, alerts, rule-based profitability advisor); Audience/Creative Analytics and AI-generated narrative features deferred to Phase 2.

29.2 Next Steps

1. Confirm the Manager vs. Marketing Executive permission split in Section 21 matches intent, since it is not yet fully encoded in application-layer checks beyond the workspace_members table.
2. Review and mark up this draft (section-by-section comments welcome).
3. Break the remaining MVP scope (Section 24) into sprint-sized epics for build planning.
4. Confirm WhatsApp Business API account status/lead time before committing to a launch date for that alert channel.
5. ER diagram (Section 18.2): generate as a visual artifact whenever convenient — the entity/relationship list in 18.1 is ready.